

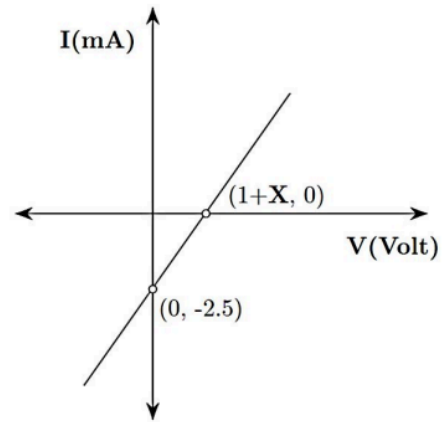
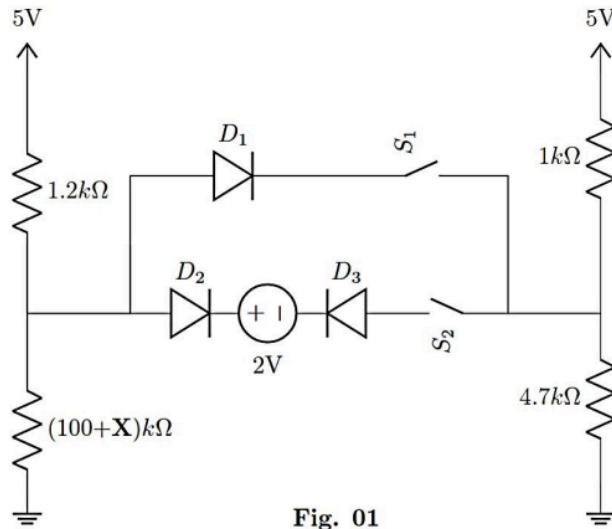
# BRAC UNIVERSITY

## Department of Computer Science and Engineering

Assignment 02, Summer 2024

CSE 251: Electronic Devices and Circuits

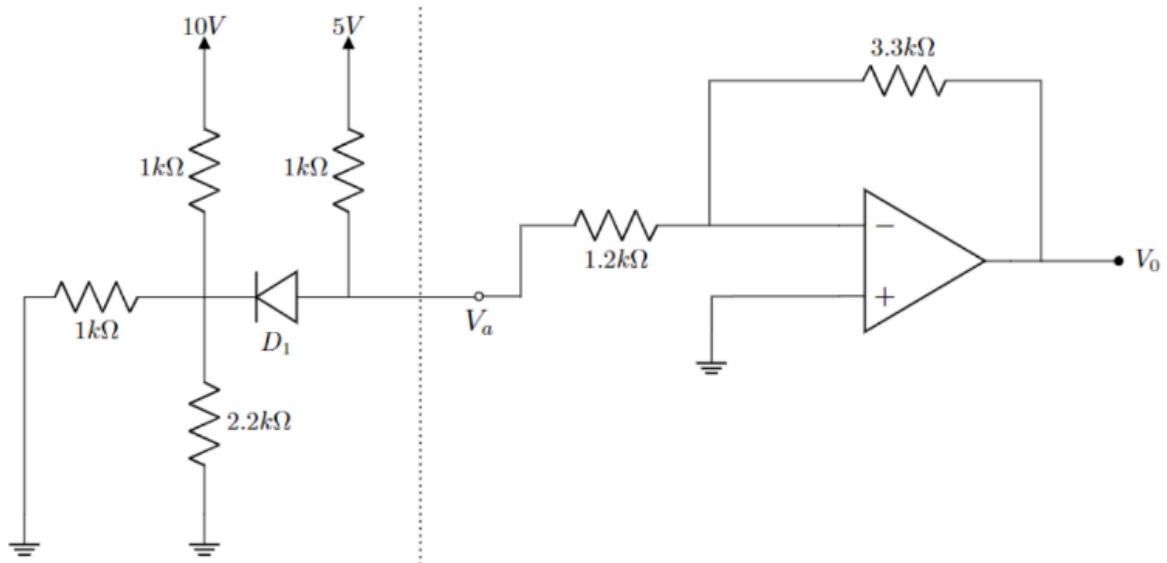
1.



In Fig. 01, a circuit is shown with two ‘off’ switches, S1 and S2. Assume X=last digit of your ID. ( $V_{D0} = 0.7V$ ).

- If only S1 is ‘on’, determine the current passing through the diode D1. [4]
- If only S2 is ‘on’, determine the current passing through the diode D2 and D3. [4]
- Find and draw the unknown component in Fig. 02. [2]

2.



As shown in the figure, a resistor-diode network is constructed and one of its nodes  $V_a$  is fed to the ideal Op-Amp to the right side. The Op-Amp has been connected to the saturation voltages  $V_{sat}^+ = +10\text{ V}$  and  $V_{sat}^- = -10\text{ V}$ . Assume that  $V_D = 0.7\text{ V}$ .

- Determine whether the diode  $D_1$  is conducting or not. [4]
- Calculate which  $1\text{ k}\Omega$  resistor (left/middle/right) dissipates the least power. [3]
- Analyze the entire circuit to find what the voltage  $V_0$  should be. [3]